

Flame Weeding For Effective Weed Control

Weed pressure can be an issue with many different crops, but 2 of them are particularly challenging: carrots, which are slow to germinate, and onions, which don't grow large canopies that would ultimately shade out weeds. For both these crops, flame weeding can prove to be a very cost efficient solution.

Stale Seedbed and Pre-emergence Flame Weeding

The stale seedbed method is a weed control technique whereby beds are prepared before the scheduled seeding date in order to grow weeds that will then be destroyed before the crop is planted. This strategy can be used for all crops, but carrots often take longer to germinate than the weed seeds, so we take this idea a step further by seeding the crop BEFORE we destroy the weeds that have sprouted in the beds. Here are the steps :

1. Start by preparing the beds as you normally would, but do so 1 week before the scheduled seeding date. Fertilize the beds, irrigate and even use row covers, following the same steps you would if you were going to seed the bed.
2. One week later, at the scheduled time of seeding, sow the bed even if some weeds have sprouted on the bed surface.
3. Simultaneously sow a couple seeds of an indicator plant (a seed that will begin germinating a little before the main crop under the same growing conditions) at one end of the bed and mark the spot with a flag. Set an alarm reminding you when your indicator plant should germinate, usually in 6 to 7 days, depending on the time of the year. For carrots the indicator plant is beets.

To use pre-emergence flame weeding with direct seeded beets, you can use radish as an indicator plant.

4. When the indicator seeds germinate, destroy the weeds with a flame weeder. This technique destroys weeds via thermal shock, without working the soil and disturbing the crops germinating underneath.

Calendar - Stale Seedbed

When?	Action!
7-10 days before seeding	Bed preparation Fertilization of the beds Irrigation of the beds Row covers
Seeding day	Seed the crop Sow a few seeds of a crop that germinates faster Set an alarm on your phone for 5 days from now
5-7 days after seeding and/or when the indicator seeds have germinated	Check that the crop has not germinated Destroy weeds with a flame weeder

Number of days required for germination of direct seeded crops, depending on soil temperature

Crop	Soil Temperature					
	50°F (10°C)	59°F (15°C)	68°F (20°C)	77°F (25°C)	86°F (30°C)	95°F (35°C)
Carrot	17 days	10 days	7 days	6 days	6 days	9 days
Beet	16 days	8 days	6 days	5 days	4.5 days	5 days
Radish	11 days	6 days	4 days	3.5 days	3 days	-

Using The Flame Weeder With Alliums

The objective of flame weeding is not to burn the weeds, but to give them a thermal shock which will kill them. The main advantage of this strategy over other weeding techniques is that the soil is not disturbed and therefore dormant weed seeds are not exposed and do not germinate.

Alliums are often referred to as “dirty” crops because even with intensive spacing, the canopy of the plant won’t cover the bed surface enough to limit light exposure, resulting in weed germination and growth.

Another characteristic of alliums is that they can endure exposure to a light thermal shock without getting damaged. When the onions reach the 5 leaves stage, weeds can be killed by using the flame weeder torch.

Keeping the beds clean during these first few stages of the crop’s growth is essential for the optimal development of the onion bulb. For flame-weeding to be successful, the flames of the torch must come into contact with the soil at a 45 degree angle, and the weeds must be small enough for the relatively brief flame exposure to kill them.

